DupeCheck: Product Requirements Document (PRD)

Field	Detail
Product	DupeCheck, duplicate test alert at order entry
Document	Product Requirements Document, v0.1 DRAFT
Author	Rohan Singh, Product Owner
Related	DupeCheck BRD, DupeCheck Project Flow
Data policy	Synthetic data only. Windows and scores marked **[Validate]** are illustrative until clinically and statistically validated

1. Product overview

Vision: every clinician knows, at the moment of ordering, whether a similar test was already done, and can decide with confidence.

Product principles

1. **Support, never block.** The clinician always decides. Urgent care is never delayed.
2. **Explain every alert.** Show why it fired and what the prior result was.
3. **One calm alert.** Quiet design, one alert per order, no flashing or sounds.
4. **Rules first.** Transparent, reviewable logic in v1. AI only where rules cannot read the data, and only after a safety gate.
5. **Measure everything.** Alerts, actions, overrides and avoided tests are logged.
6. **Fit the workflow.** Appear inside the existing ordering flow, not in a separate tool.

2. Problem and opportunity (summary)

Clinicians cannot easily see tests done at other facilities, so avoidable duplicates are ordered, adding cost, patient burden and delay. Published studies show data sharing reduces duplicate imaging, but a push warning at the point of ordering is not routine. Full evidence, sizing model and caveats are in the BRD.

3. Users and personas

Persona 1: Dr. Priya Raman, ordering clinician (primary)

- Primary care or hospitalist, sees 20 or more patients a day.
- **Jobs to be done:** decide the next test quickly; avoid repeating tests; not miss relevant history.
- **Pains:** limited time, scattered records, alert fatigue.
- **Needs:** a short, accurate, explainable alert she can act on in seconds.

Persona 2: Marcus Lee, quality and patient safety lead

- **Jobs to be done:** keep alerts useful and safe, review overrides, report outcomes.
- **Needs:** override reasons, acceptance rates, ability to tune rules under governance.

Persona 3: Anita Shah, analytics and utilization lead

- **Jobs to be done:** quantify avoided tests and spend, find patterns.
- **Needs:** reliable metrics, drill down by test type and site, defensible assumptions.

Persona 4: Compliance officer

- **Needs:** audit log of who saw external data and why, role based access evidence.

4. Goals, non-goals and success metrics

Goals (v1)

- Detect likely duplicate tests at order entry for selected test types.
- Show a clear, explainable alert with the prior test and source.
- Capture clinician action and reason.
- Provide measurement for leadership.

Non-goals (v1): automated decisions, denying or blocking orders, patient facing features, claims adjudication, predictive models, free text report reading, replacing any EHR.

Metrics tree

Level	Metric	Definition	Target (illustrative)
North star	Avoidable duplicates avoided per 1,000 eligible orders	Orders cancelled or modified after alert where no clinical reason to repeat	Baseline in Phase 0, then 20 to 30 percent reduction
Adoption	Alert action rate	(Cancelled + modified) / alerts shown	At or above 30 percent
Quality	Override rate and reasons	Overrides / alerts shown, split by reason	Reviewed monthly, trend down for "no reason given"
Trust	False alert rate	Alerts judged wrong by clinicians / alerts shown	Under 10 percent in pilot
Speed	Alert latency	Time from order select to alert	Under 2 seconds at 95th percentile
Safety	Blocked urgent orders	Count	Zero
Value	Estimated avoided cost	Avoided tests x average cost	Validated with finance
Satisfaction	Clinician rating	Pilot survey	At or above 4 out of 5

5. Scope by release

Release	Scope	Gate to next
Phase 0	Discovery, data baseline, rule windows proposed, clinical advisory formed	Duplicate rate and avoidable share justify build
MVP (v1)	Rules engine, order time alert, prior result view, override reasons, audit log, admin insights, 5 to 8 test types	Pilot metrics meet targets, safety review passed
v2	More test types and sites, claims data feed, rule management console, scheduled reports	Sustained action rate, low false alerts
v3 (gated)	NLP on free text reports, test mapping with embeddings, smarter alerting, anomaly insights	Clinical safety and validation approval

6. User journey (primary)

1. Clinician selects a patient and starts ordering a test.
2. On selecting the test, DupeCheck checks recent results across available sources.

3. **No match:** nothing appears, order proceeds.
4. **Match outside window:** a small information note appears inline.
5. **Match inside window:** one calm alert opens with prior test, date, facility, result summary and reason.
6. Clinician chooses: cancel and use prior result, order a modified test, or order anyway with a reason.
7. The outcome is logged and appears in admin insights.

7. Feature list and prioritization

MoSCoW with an illustrative RICE style score (Reach x Impact x Confidence / Effort). **Scores are placeholders to be validated [Validate].**

ID	Feature	MoSCoW	Reach	Impact	Confidence	Effort	Score
F1	Duplicate detection rules engine (test code, modality plus body region, window)	Must	10	3	0.8	5	4.8
F2	Order time alert with explanation	Must	10	3	0.8	3	8.0
F3	Prior result preview	Must	8	2	0.8	3	4.3
F4	Clinician actions with override reasons	Must	10	2	0.9	2	9.0
F5	Audit log	Must	10	2	1.0	2	10.0
F6	Admin insights dashboard	Must	5	2	0.8	3	2.7
F7	Role based access	Must	10	2	1.0	2	10.0
F8	Info note for matches outside window	Should	6	1	0.8	1	4.8
F9	Rule management console (governed edits)	Should	4	2	0.7	4	1.4
F10	Claims data feed for external tests	Should	8	3	0.5	6	2.0
F11	Scheduled reports and anomaly flags	Could	3	1	0.6	3	0.6
F12	NLP on free text reports	Could (v3)	8	3	0.4	8	1.2
F13	Embedding based test mapping	Could (v3)	8	2	0.5	6	1.3
F14	Patient facing notifications	Won't (v1)					
F15	Hard stop or order denial	Won't (never)					

Prioritization note: F10 scores low on confidence because claims latency and coverage are unproven, so it is a v2 feature after Phase 0 data profiling.

8. User stories and acceptance criteria

US-01 Duplicate alert (F1, F2) As an ordering clinician, I want to be warned when a similar test was recently done so that I avoid repeating it unnecessarily.

- Given a patient with a CT chest 21 days ago and a 90 day window, when I select CT chest, then one alert appears within 2 seconds.
- The alert shows prior test name, date, "days ago", source facility and a one line result summary.
- The alert states the reason, for example "Same test and body area within the 90 day review window".
- If no prior test exists, no alert appears.

US-02 External results included (F1, F3) As an ordering clinician, I want tests from other facilities included so that I see the full history.

- External results are tagged "External" with the facility name.
- If the source cannot be reached, the order proceeds and a small notice says external results were unavailable.

US-03 Prior result preview (F3) As an ordering clinician, I want to view the prior report quickly so that I can judge whether repeating is justified.

- "View prior report" shows impression and key findings in the same screen, without leaving the order.

US-04 Act on the alert (F4) As an ordering clinician, I want clear options so that I can decide fast.

- Options: cancel and use prior result, order a modified test, order anyway.
- "Order anyway" requires a reason from a list (clinical change, inadequate prior result, poor image quality, worsening condition, other with free text).
- The text "Final decision rests with the clinician" is visible.

US-05 Urgent orders never blocked (F2, F4) As an ordering clinician, I want urgent orders never delayed so that care is not harmed.

- For urgent priority, the alert is shown but "Order anyway" is pre-emphasized and needs no extra step beyond selecting a reason.
- No path in the product prevents an order from being placed.

US-06 Information note outside window (F8) As an ordering clinician, I want a light note when a prior test exists but is outside the window so that I stay informed without interruption.

- A small inline note appears, no modal.

US-07 Audit trail (F5) As a compliance officer, I want every alert and decision logged so that I can audit use.

- Each event stores timestamp, user, patient reference, test, outcome, reason and data sources viewed.
- Logs are read only and exportable by authorized roles.

US-08 Insights (F6) As an analytics lead, I want to see alert and outcome metrics so that I can prove value.

- Shows alerts shown, actions, overrides, avoided tests and estimated avoided cost.
- Filter by date, test type and site.
- Cost assumptions are visible and labelled illustrative until validated.

US-09 Governed rules (F9) As a quality lead, I want to adjust review windows under governance so that alerts stay relevant.

- Changes require an approver, are versioned and take effect only after approval.

US-10 Access control (F7) As a compliance officer, I want role based access so that only authorized users see sensitive data.

- Roles: clinician, quality lead, analyst, compliance, admin.
- Clinicians see only their patients' data; analysts see de-identified aggregates by default.

9. Alert design requirements

- One alert per order. No stacking of multiple warnings.
- Calm styling: amber for caution, no red unless destructive, no sound, no animation.
- Plain language reason, not a score.
- Alert can be dismissed with one action, and the reason requirement appears only for "order anyway".
- Reading time target under 10 seconds.
- Accessible: keyboard navigable, screen reader labels, sufficient contrast.

10. Business rules (illustrative [Validate] with clinical advisory)

Test	Window
CT and MRI	90 days
X-ray	30 days
HbA1c	60 days
Lipid panel	30 days
CBC	14 days
Echocardiogram	180 days

- Match logic: same standard test code, or same modality plus body region.
- External results count as matches.
- Windows are configurable by clinical governance and versioned.
- Clinical exceptions (for example follow-up imaging explicitly required by protocol) are handled by an exception list reviewed by governance.

11. Non-functional requirements

Area	Requirement
Performance	Alert in under 2 seconds at 95th percentile
Availability	If DupeCheck is down, ordering continues unaffected
Security	Encryption in transit and at rest, role based access, audit logging, HIPAA minimum necessary
Privacy	Synthetic data in non production; PHI only in approved environments with a BAA
Interoperability	HL7 FHIR R4 resources (Patient, Observation, DiagnosticReport, ServiceRequest, ImagingStudy); LOINC and CPT mapping; CDS Hooks style integration to be confirmed with architecture
Explainability	Every alert includes the rule that fired
Maintainability	Rules configurable without code release

12. Analytics and event tracking

Events to log: order_started, alert_shown, alert_viewed_report, order_cancelled, order_modified, order_overridden (with reason), info_note_shown, source_unavailable. Each event carries timestamp, test type, site, user role and alert id. No free text clinical content in analytics events.

13. Rollout and pilot plan

1. Phase 0 baseline using de-identified data.
2. Shadow mode: compute alerts silently, compare with actual orders to estimate precision without affecting care.
3. Limited pilot with a small group of volunteer clinicians at one site and one or two test types.
4. Review weekly: action rate, false alerts, feedback.
5. Expand test types and sites only after targets are met and safety review is complete.

14. Dependencies

EHR integration capability, access to external results and claims feeds, patient matching service, clinical governance for rules, compliance approval, data engineering capacity.

15. Risks and open questions

Key risks are in the BRD. Product specific open questions:

1. Which tests make the best first wave (high cost, frequent repeats, clear windows)?
2. Should the alert appear at order select or at order sign, or both?
3. Who owns the exception list?
4. What claims latency is acceptable for the claims feed to be useful?
5. How is clinician feedback captured without adding clicks?

16. Traceability to BRD

Business requirement	Features	Stories
BR-01 Baseline	Phase 0 analysis, F6	US-08
BR-02 Warn at order time	F1, F2	US-01, US-02
BR-03 Explain the alert	F2, F3	US-01, US-03
BR-04 Never block	F2, F4	US-04, US-05
BR-05 Override reasons and audit	F4, F5	US-04, US-07
BR-06 Tunable rules	F9	US-09
BR-07 Leadership visibility	F6	US-08
BR-08 Access and audit	F5, F7	US-07, US-10
BR-09 Claims plus clinical data	F10	US-02
BR-10 AI ready design	F12, F13	Future

17. Prototype reference

A clickable front end prototype (React and Tailwind, synthetic data) demonstrates three scenarios: a strong duplicate, a match outside the window, and no prior test. It covers Order Entry, Patient Timeline and Admin Insights and should be linked here when published.